WORKED SOLUTION

STEP III 2013, Question 10

Mechanics • Rotational Impact

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Read the question before you go on. It is in the STEP Support Programme's question database:

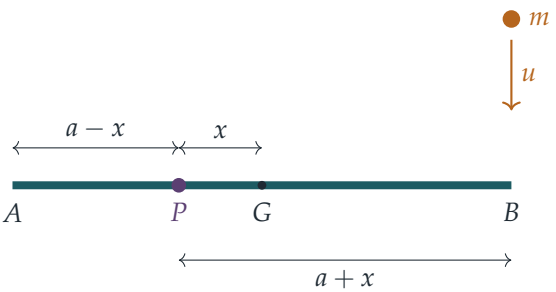
step.maths.org/questions/13-s3-q10

The key idea

The moment of inertia comes straight from the parallel axis theorem, once we notice how far the centre of the rod is from P. For the impact, the axis exerts an unknown impulse on the rod, so linear momentum is *not* conserved; but that impulse acts through P, so it has no moment about P. Conservation of angular momentum about P, together with Newton's experimental law at B, gives two equations for the two unknowns.

Solution

We model the rod as a uniform rigid rod, the particle as a particle, and the impact as instantaneous, so that the only horizontal impulses are those at B and at the axis (finite forces such as gravity give no impulse during the instant of impact).



The moment of inertia

Let G be the midpoint of the rod, its centre of mass. Since $AG = a$ and $AP = a - x$, the distance PG is x (with G on the B side of P when $x > 0$). The moment of inertia of a uniform rod about a perpendicular axis through its centre is $\frac{1}{3}Ma^2$ (for a rod of length $2a$). By the parallel axis theorem,

$$I_P = \tfrac{1}{3}Ma^2 + Mx^2 = \tfrac{1}{3}M(a^2 + 3x^2),$$

as required.

If you prefer to work from first principles: with line density $M/(2a)$ and s measured along the rod from P, the rod runs from $s = -(a-x)$ to $s = a+x$, and

$$I_P = \frac{M}{2a}\int_{-(a-x)}^{a+x} s^2\,\mathrm{d}s = \frac{M}{6a}[(a+x)^3 + (a-x)^3] = \frac{M}{6a}(2a^3 + 6ax^2),$$

which is the same thing.

The impact

The distance from P to B is $2a - (a-x) = a+x$. Let v be the particle's velocity immediately after impact (in the original direction of motion) and ω the rod's angular velocity, so that B then moves with speed $(a+x)\omega$ in the same direction.

Ask yourself Why can't we use conservation of linear momentum here?

Because the axis is fixed, it supplies an impulse on the rod during the collision, of unknown size. That impulse acts at P, so its moment about P is zero. The impulse between the particle and the rod is internal to the system of rod and particle. So the total angular momentum of the system about P is conserved:

$$mu(a+x) = mv(a+x) + I_P\,\omega. \tag{1}$$

Newton's experimental law at B (speed of separation is e times speed of approach) gives

$$(a+x)\omega - v = eu. \tag{2}$$

From (2), $v = (a+x)\omega - eu$. Substituting into (1),

$$\begin{aligned} mu(a+x) &= m(a+x)[(a+x)\omega - eu] + I_P\,\omega \\ mu(1+e)(a+x) &= [m(a+x)^2 + \tfrac{1}{3}M(a^2+3x^2)]\omega. \end{aligned}$$

Multiplying through by 3,

$$\omega = \frac{3mu(1+e)(a+x)}{M(a^2+3x^2) + 3m(a+x)^2},$$

which is the given result.

The case $m = 2M$

Put $m = 2M$ and cancel M:

$$\omega = \frac{6u(1+e)(a+x)}{a^2+3x^2+6(a+x)^2}.$$

The variable that appears naturally is the lever arm $s = a+x$, the distance PB, which lies in $[0, 2a]$. With $x = s-a$, the denominator is

$$a^2 + 3(s-a)^2 + 6s^2 = 9s^2 - 6as + 4a^2,$$

so

$$\omega = \frac{6u(1+e)\,s}{9s^2 - 6as + 4a^2}.$$

At $s = 0$ (the axis at B) we have $\omega = 0$, so the maximum occurs with $s > 0$, and there we can divide top and bottom by s:

$$\omega = \frac{6u(1+e)}{9s + \dfrac{4a^2}{s} - 6a}.$$

We want the denominator as small as possible. Since $s > 0$, the AM–GM inequality gives

$$9s + \frac{4a^2}{s} \;\geq\; 2\sqrt{9s \cdot \frac{4a^2}{s}} = 12a,$$

with equality exactly when $9s = 4a^2/s$, that is $s = \frac{2}{3}a$, which does lie in $(0, 2a]$. So the denominator is at least $12a - 6a = 6a$, with equality only at $s = \frac{2}{3}a$, and so

$$\omega \leq \frac{6u(1+e)}{6a} = \frac{u(1+e)}{a}.$$

The greatest angular velocity is therefore $u(1+e)/a$, attained when $a + x = \frac{2}{3}a$, that is

$$x = -\tfrac{1}{3}a.$$

The negative sign just means that P is on the B side of the centre: P is a distance $\frac{4}{3}a$ from A and $\frac{2}{3}a$ from B.

Checks

- At $x = a$ (the axis at A) with $e = 1$ the formula gives $\omega = 3mu/\big(a(M+3m)\big)$; letting $M \to 0$ gives $\omega = u/a$, so B moves at $2u$, which is the familiar result for a very light "target" struck elastically.
- In the optimal case with $a = 1$, $u = 1$, $e = \frac{1}{2}$, the rod turns at $\omega = 1.5$, so B moves at 1, and the particle's velocity after impact is $v = 1 - \frac{1}{2} = \frac{1}{2}$. Then the angular momentum about P is $2M \cdot \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{3}M\big(1 + \frac{1}{3}\big) \cdot 1.5 = \frac{4}{3}M$, matching $2M \cdot 1 \cdot \frac{2}{3}$ before impact.
- A fine search by computer over $x \in [-a, a]$ confirms the maximum at $x = -a/3$.

Common slips

- Conserving linear momentum: the fixed axis gives an impulse, so only angular momentum about P is conserved.
- Taking the distance PB as $a - x$ or $2a - x$ rather than $a + x$.
- Rejecting $x = -\frac{1}{3}a$ because it is negative: x is a signed position, and any x in $[-a, a]$ puts P on the rod.

Beyond STEP

Choosing the point about which to take moments so that an unknown reaction drops out is one of the most useful tricks in rigid-body dynamics; it reappears in the "centre of percussion", the point of a bat or rod at which a blow produces no impulse at the pivot. First-year mechanics develops this into the general theory of angular momentum and the inertia tensor. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats impulsive rotation and the centre of percussion very clearly.

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